

SUMUKH ACHARYA

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SKILLS

Languages: Python, JavaScript, MySQL

Frameworks and Tools: FastAPI, Flask, Docker, Git, Kafka, Gunicorn, Uvicorn

ML/AI: PyTorch, LoRA, Transformers, Fine-Tuning, Streamlit, HuggingFace

EXPERIENCE

AI Intern

tvam Technologies

Sep 2025 - Present

Bengaluru, India

- Fine-tuned a 270M parameter SLM using LoRA/QLoRA, achieving benchmark scores within 15 points of GPT-5.2 on structured output tasks across 2 full adaptation cycles.
- Self-curated a 1.5M token instruction dataset including prompt templates and edge-case scenarios for instruction tuned and supervised fine-tuning pipelines.
- Conducted cross-model benchmarking across foundation and instruction-tuned models, measuring output quality, latency trade-offs, and behavioral consistency.
- Built and containerized a full-stack inference app (FastAPI, JavaScript, Docker) serving the fine-tuned model via a chat UI, collecting user interaction logs as preference data for downstream DPO alignment.
- Completed parallel training in AI fundamentals, app dev, system design, Python, and DSA to support production-level fine-tuning and deployment decisions.

Teaching Assistant

PES University | Python, Databases, Linear Algebra

Aug 2025 - Present

Bengaluru, India

- Supported 150+ students across Python and Databases courses through lab sessions, assignment grading, and one-on-one debugging and doubt-clearing.
- Prepared practice problems and lab material covering Python and Databases fundamentals.
- Currently assisting Linear Algebra course (Jan 2026 – May 2026).

Data Science Intern

CODMAV Research Centre (PES University)

Jun 2024 - Aug 2024

Bengaluru, India

- Built a [lung cancer risk prediction system](#) on 22,811 patient records, achieving 98.75% accuracy and 96.25% recall using a soft-voting ensemble of XGBoost, CatBoost, SVM, and KNN on RFE-selected features.
- Engineered a multi-stage data pipeline reducing 788 clinical features to 89 medically-validated markers via RF imputation and data filtering.
- Compared 4 feature selection algorithms (PCA, BSO, RFE, SelectKBest) across multiple model architectures to identify the optimal clinical feature subset.
- Published findings at IEEE InC4 2025, indexed in IEEE Xplore.

PROJECTS

Grid Pulse — Real-Time F1 Stock Market | Python, Apache Kafka, Streamlit

- Built a real-time data pipeline simulating a stock market for F1 drivers, processing both historical batch data and live race weekend streams via Kafka.
- Designed a distributed architecture decoupling data generation, processing, and visualization with cron-triggered real-time updates and a Streamlit dashboard.

DFOS — Distributed File Orchestration and Synchronization | Python, Sockets, Threading

- Built a multi-client file transfer system with concurrent client handling using ThreadPoolExecutor, supporting upload, download, preview, and delete operations.
- Implemented user authentication, dedicated per-user storage, and performance logging with CPU/memory monitoring on shutdown.

Senior Vision AI — Scan-n-Say | Python, Flask, Google Vision API, Gemini API

- Built an accessible mobile web app for elderly and visually impaired users that scans supermarket products and reads out AI-generated summaries aloud.
- Integrated Google Cloud Vision (OCR) and Gemini API for text extraction and summarization; deployed backend on Render and frontend on Vercel.

URL Shortener | Python, Flask, Docker, Kubernetes, Redis

- Built a production-ready URL shortening service with Kubernetes deployment, horizontal pod autoscaling, NGINX load balancing, and Redis caching.
- Stress tested at 1,000 concurrent requests achieving 88 req/sec with 100% success rate across all pods.

PUBLICATIONS

Acharya, S., et al. **Predictive Analytics for Early Lung Cancer Risk using Machine Learning.** IEEE International Conference on Contemporary Computing and Communications (InC4), 2025. Indexed in IEEE Xplore.

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **7th out of 364 teams** in [Kaggle Data Analytics Hackathon](#).
- [IBM Data Science Professional Certificate](#) — Coursera.

EDUCATION

B.Tech in Computer Science and Engineering

Expected: May 2026

PES University, Bengaluru | Specialisation: Machine Intelligence and Data Science

CGPA: 7.51

Relevant Coursework: Machine Learning, Deep Learning, Databases, Big Data, Statistics for Data Science, Linear Algebra